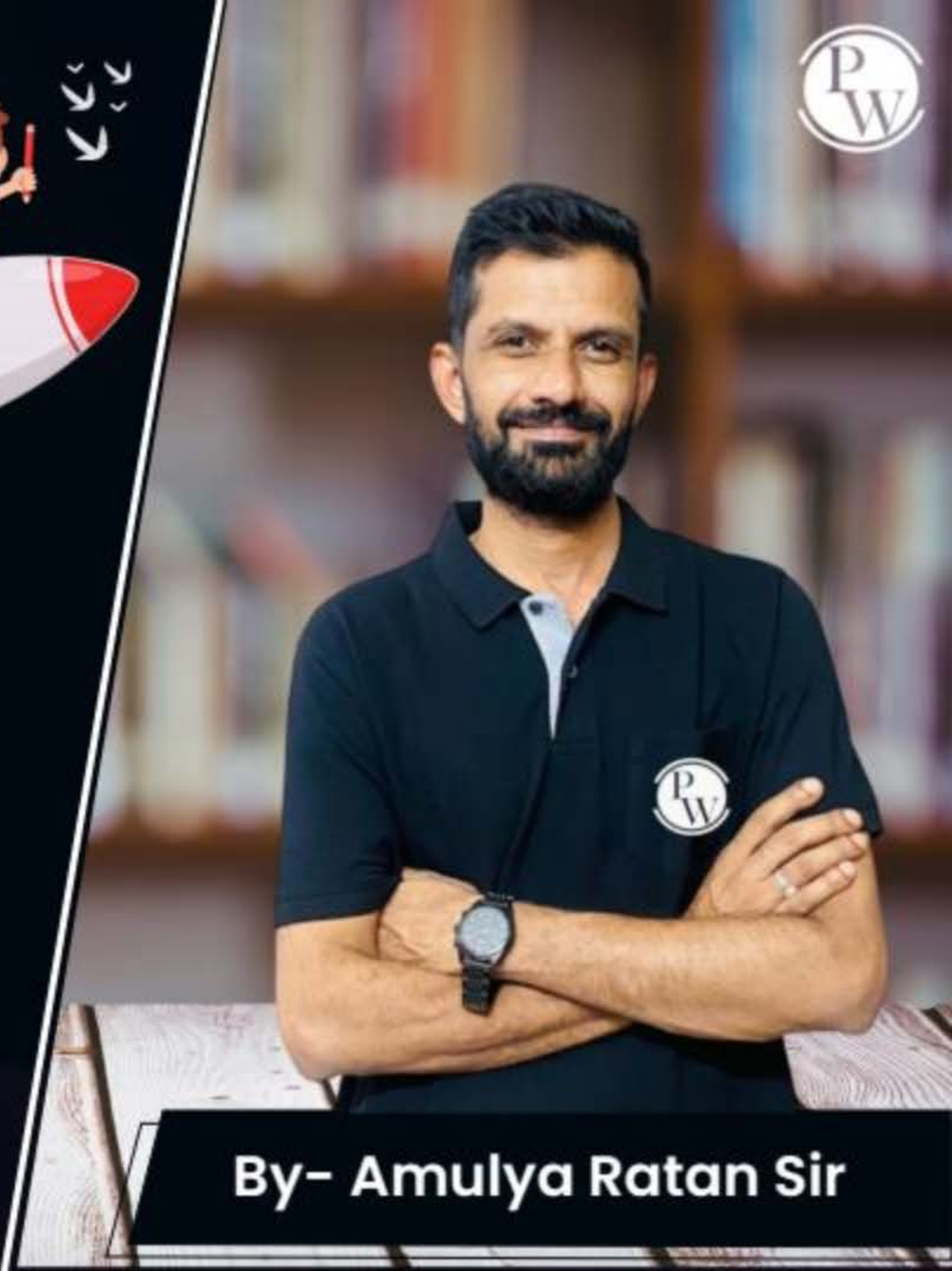
GATE ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 13



By- Amulya Ratan Sir

Recap of Previous Lecture



Topic Pipes and Cistern, Basics of Time & Distance

Topics to be Covered

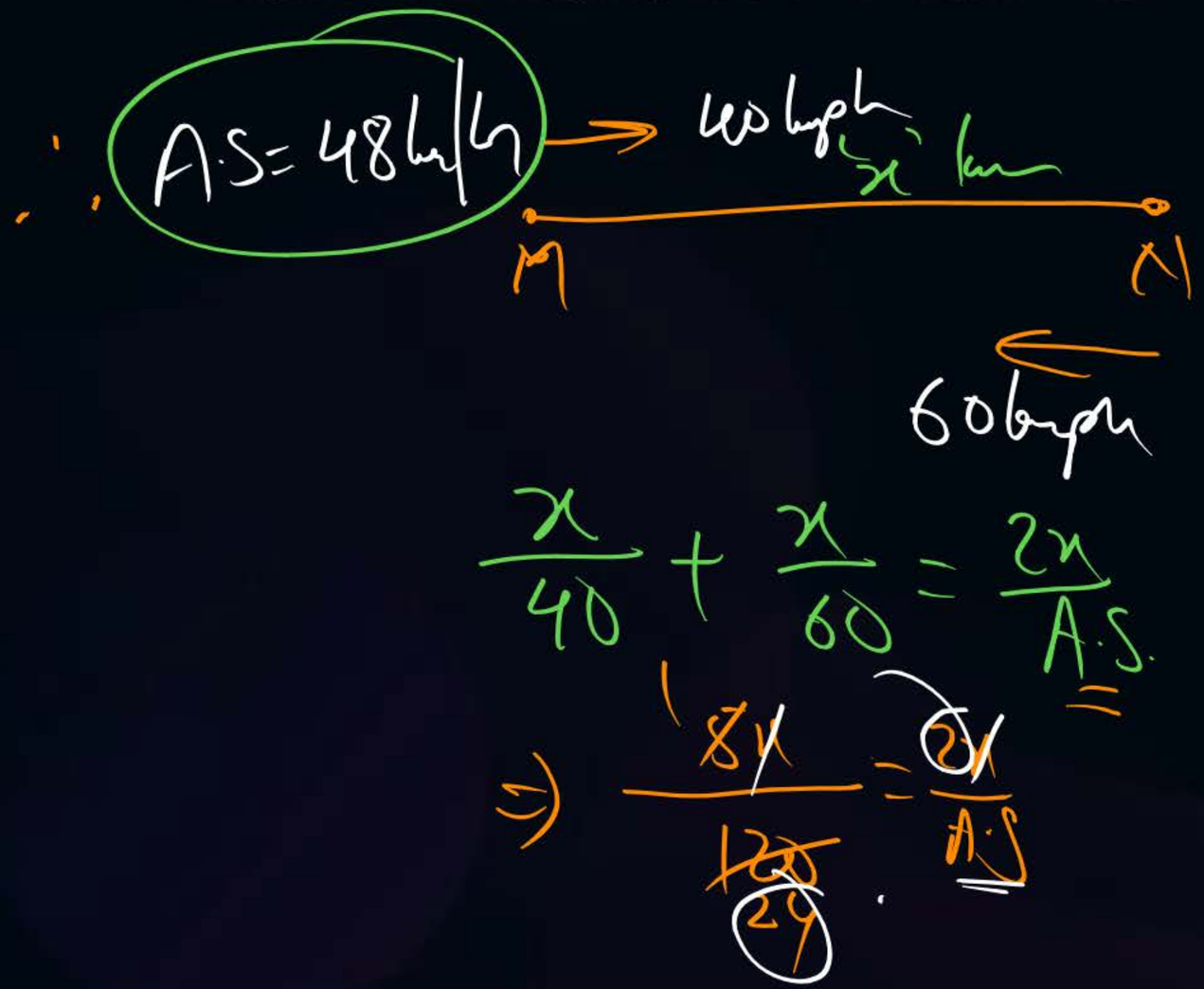


Topic-1 Time and Distance

[MCQ]

#Q. If you travel from M to N at 40km/hr and N to M at 60 km/hr. What would be your average speed of the journey?

Assignment



[MCQ]

#Q. A man travels from Howrah to Kanpur in his car. If he covers 1/3rd of the distance at 60 km/hr & remaining at 40 km/hr. Find the average speed of his journey.



$$\frac{x}{60} + \frac{2x}{40} = \frac{3x}{A.S.}$$

$$\Rightarrow \frac{2x + 6x}{120} = \frac{3x}{A.S.}$$

$$\frac{8x}{120} = \frac{3x}{A.S.}$$

$$\frac{1}{15} = \frac{3}{A.S.}$$

$$A.S. = 45 \text{ km/hr}$$

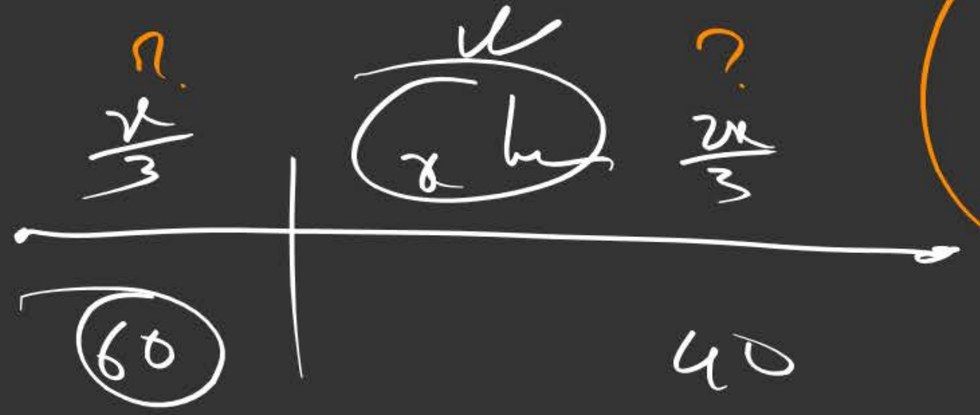
Assignment

Average speed = Total Distance / Total Time

$$A.S. \times T.T. = \text{Total Dist}$$

$$T.T. = \frac{T.P.}{A.S.}$$

$$\frac{x}{60} + \frac{2x}{40} = \frac{3x}{A.S}$$



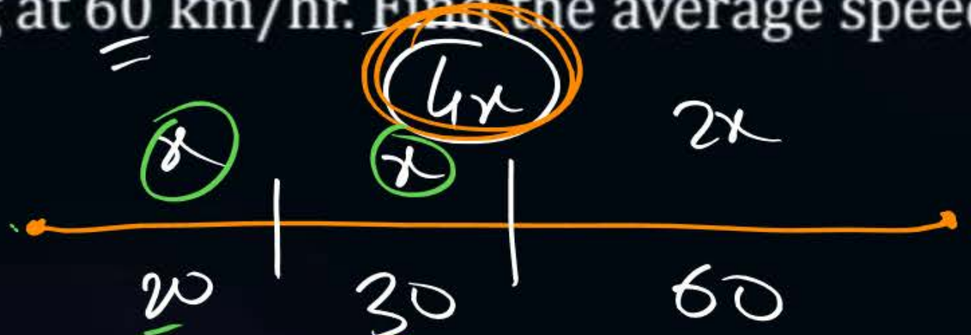
$$\frac{2xy}{x+y}$$

$$\frac{x}{3} \times 60 + \frac{2x}{3} \times 40 = \frac{x}{A.S.}$$

$$\frac{1}{3}\left(\frac{x}{60} + \frac{2x}{40}\right) = \frac{x}{A.S}$$

[MCQ]

#Q. A man travels ¼th of his journey at 20 km/hr, another ¼th at 30 km/hr & remaining at 60 km/hr. Find the average speed of his journey.



$$\frac{x}{20}+\frac{x}{30}+\frac{2x}{60}=\frac{4x}{A.S.}$$

$$\Rightarrow \frac{7x}{60}=\frac{4x}{A.S.}$$

$$A.S.=\frac{60\times4}{7}$$

$$=\frac{240}{7}$$

$$=34\frac{2}{7}$$

$$34.28$$

Assignment

$$\frac{2\times20\times30}{50}=24\ km/hr$$

$$\frac{2\times24\times60}{84}=\frac{240}{7}$$

$$=34.28$$

[MCQ]

#Q. If a car travels along four sides of a square at 100 kmph, 200 kmph, 300 kmph and 400 kmph. Find its average speed.

A.S. = 48 × 4 = 192

A 152 km/hr

B 180 km/hr

C 192 km/hr

D 210 km/hr

$$\frac{x}{100}+\frac{x}{200}+\frac{x}{300}+\frac{x}{400}=\frac{4x}{A.S}$$

$$\Rightarrow \frac{12x+6x+4x+3x}{1200}=\frac{4x}{A.S}$$

$$\Rightarrow \frac{25x}{1200}=\frac{4x}{A.S}$$

[MCQ]

#Q. A tourist covers half of his journey by train at 60 km/hr, half of the remainder by bus at 30 km/hr and the rest by cycle at 10 km/hr. Average speed of the tourist during the journey is

A. 36 km/hr

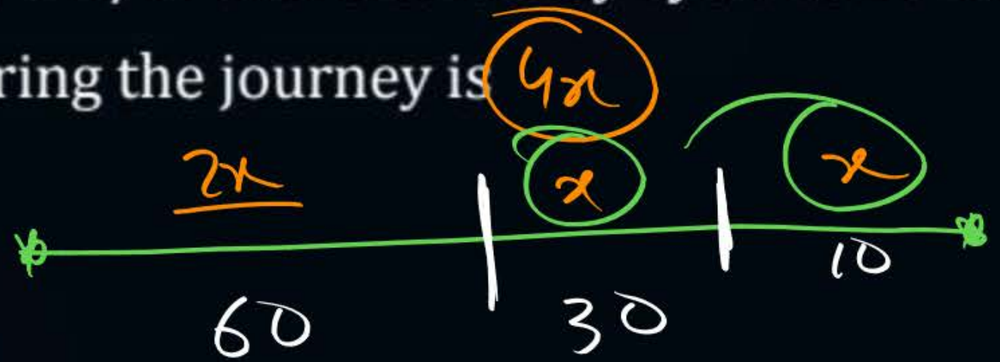
B. 33 km/hr

C. 24 km/hr

D. 18 km/hr

$\frac{10x}{60} = \frac{4x}{A.S}$

$A.S = 24 km/hr$



$\frac{2x}{60} + \frac{x}{30} + \frac{x}{10} = \frac{4x}{A.S}$

$\Rightarrow \frac{2x+2x+6x}{60} = \frac{4x}{A.S}$

[MCQ]

#Q. A person covers a certain distance at a certain speed. If he decrease his speed by 20%, then he takes 16 minutes more to cover the same distance. Find the time taken by him to cover the original distance at original speed.

$U \cdot T = 64$ min

| 5 : 4

$\frac{100}{80}$

$U \cdot T - U \cdot \frac{4}{5}T = 16$ min

$\left(\frac{5}{4} - 1\right) U \cdot T = 16$ min

$\frac{1}{4} U \cdot T = 16$ min

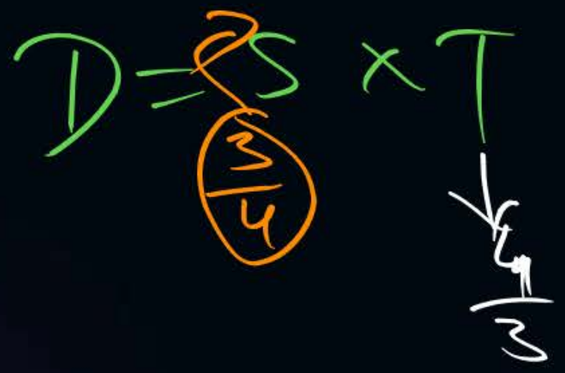
D = S × T ?

$\frac{80}{100}$

$\frac{4}{5}$

[MCQ]

#Q. Walking 3/4th of the usual speed a man is late to office by 10 minutes, find the usual time.



A 40 minutes

B 12 minutes

C 60 minutes

D 30 minutes

$\frac{4}{3}S \cdot T - S \cdot T = 10\text{min}$

$\left(\frac{4}{3} - 1\right) S \cdot T = 10\text{min}$

$\frac{1}{3} S \cdot T = 10$

$S \cdot T = 30\text{min}$

[MCQ]

#Q. Mumbai police observes a thief 800 metres away from him. The thief started running at 80 km/hr & the police at 170 km/hr. In how much time the police can catch the thief?

A. 8.88 minutes

B. 8.88 hours

C. 32 seconds

D. 32 minutes

$D = 800\ m$

$R.S. = 90\ km/hr \times \frac{5}{18} = 25\ m/sec$

$T = \frac{800\ m}{25\ m/sec} = 32\ seconds$

[MCQ]

#Q. An employee goes to his office from his house at a speed of 60 km/hr and reaches his office 10 minutes late. If he follows the speed of 80 km/hr, he reaches his office 5 minutes early. Find the distance he covers to reach his office.

A 55 km

B 45 km

C 50 km

D 60 km

$D = S \times T$

$T = \frac{D}{S}$

10:00 am

$60\left(T + \frac{10}{60}\right) = 80\left(T - \frac{5}{60}\right)$

$T = \frac{5}{6}$

$D = 60$ km

$x = 60$

$\frac{x}{60} - \frac{x}{80} = \frac{15}{60}$ 1

$\Rightarrow \frac{x}{240} = \frac{1}{4}$

[MCQ]

#Q. A man misses a train by 40 minutes if he travels at 30 kmph. If he travels at 40 kmph, then also he misses the train by 10minutes. What is the minimum speed required to catch the train on time?

Assignment

A 50 km/hr

B 55 km/hr

C 60 km/hr

D 45 km/hr

[MCQ]

#Q. A car driver driving at a speed of 68 km/hr locates a lorry 40 meters ahead of him. After 10 seconds, the lorry is 60 meters behind him. Find the speed of the lorry.

Assignment

[MCQ]

#Q. Two boys begin together to write out a booklet containing 817 lines. The first boy starts with first line, writing at the rate of 200 lines an hour and the second boy starts with last line. He writes line 817 and so on backwards proceeding at the rate of 150 lines an hour. At what line will they meet?

Assignment

A 469th

B 467th

C 465th

D 463rd



2 mins Summary

Topic Time & Distance

Average Speed

Relative Speed

THANK - YOU